Qian 2020 NSCLC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

6/8 samples (≥100 cells) intermix on one continuous manifold with no per-patient islands (radius-graph: 0 detached; iLISI median 1.00→2.39 of 6); both lobes (Body 3,848 / Cycling 1,400) carry the trajectory.

Gate 2 — module geography → GO (non-branching)

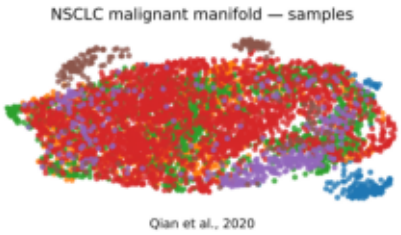
Single dominant pSom→Crest axis: pSom high over the low-cycling body declines along pseudotime (ρ=−0.68); Crest anti-correlated (r=−0.89), rising to a terminal readout (ρ=+0.82; Crest-high decile mean pt=0.98). Gast weakly structured (early programme, spent); joint pSom&Crest-high depleted (enrichment 0.18) — an inverse gradient, not a co-expression hub.

Gate 3 — hub mutual exclusivity → NA

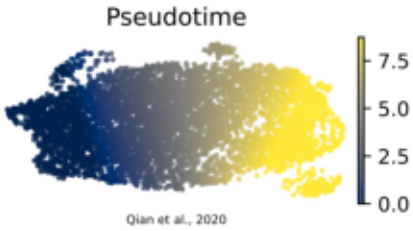
Single monotone axis — no bifurcation and no Mixed-type hub, so there is no branch pair to test for mutual exclusivity.

STEP FIGURES

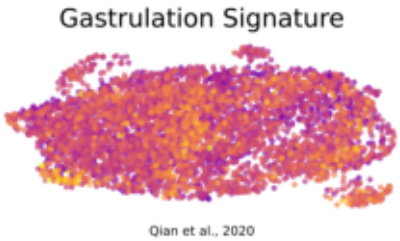
Manifold — samples



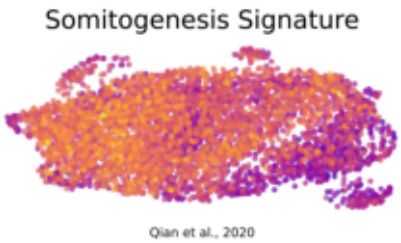
Pseudotime



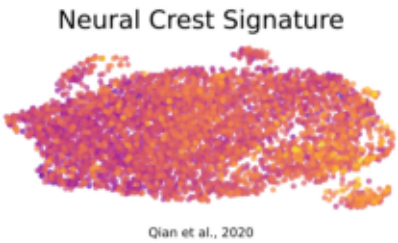
Gastrulation



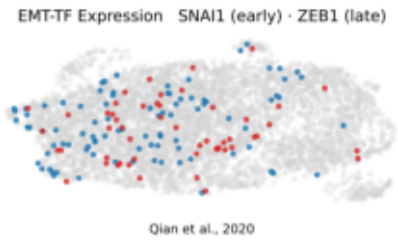
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.

Non-Small-Cell Lung Cancer (Qian et al., 2020), 10x — 8 patients / 27,262 cells (master scope); malignant only: 5,248 cells across 6 malignant-bearing samples (drop 2 sub-100 spillover samples) enter preprocessing; the whole malignant manifold (no lobe scoping) enters the trajectory.

Verdict.

Non-branching pattern reproduces — Gate 1 GO, Gate 2 GO, Gate 3 NA. Qian's NSCLC malignant manifold is a single monotone somitogenesis → terminal-neural-crest gradient (low-cycling body → neural-crest-like terminus), matching the developmental axis; it lacks Puram's hub-and-spoke bifurcation, so Gate 3 has no hub to test. The Body/Cycling split is partly proliferative and the Crest terminus enriches the cycling side, but the developmental modules are cell-cycle-free (Gast 0/81, Crest 0/104 cell-cycle genes) and the pSom↔Crest anti-correlation survives within non-cycling cells alone (r=−0.89→−0.90), so the axis is developmental, not proliferative — the whole manifold is kept (Lin/Steele no-scoping precedent), carrying pseudotime↔cycling co-variation (ρ=+0.45) as a caveat. EMT-TFs are root/body-loaded: SNAI1 (early) → ZEB1 (late) chosen by data over Puram's pair, but all expressed factors decline along pt and end-separation is weak (Δpt=0.08); the only terminus-rising factor (SNAI2, ρ=+0.18) sits just below the 3% detection floor.

Parameter choices

Min cells/sample	≥100 floor (6/8; gap non-dominant)	**Integration**	Harmony θ=10; PCA50; spread=1.3
Scope	whole manifold (no scoping)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	tip 2 = pSom-high, Crest-low	**Compartments**	Body 3,848 / Cycling 1,400